

Roll no: **43221**

Title: **Assignment 1: Study of Deep learning Packages: Tensorflow, Keras, Theano and PyTorch. Document the distinct features and functionality of the packages.**

```
In [ ]: import numpy as np
```

1. Tensorflow

```
In [ ]: import tensorflow as tf
```

```
In [ ]: print(tf.__version__)
```

2.10.0

2. Keras

```
In [ ]: from keras import datasets
        # Load MNIST datasets from keras
        (train_images, train_labels), (test_images, test_labels) =
        datasets.mnist.load_data()
```

```
In [ ]: train_images.shape
```

Out[26]: (60000, 28, 28)

```
In [ ]: test_images.shape
```

Out[27]: (10000, 28, 28)

3. Theano

```
In [ ]: !pip install Theano
```

Requirement already satisfied: Theano in ./anaconda3/lib/python3.8/site-packages (1.0.5)

Requirement already satisfied: numpy $\geq$ 1.9.1 in ./anaconda3/lib/python3.8/site-packages (from Theano) (1.20.1)

Requirement already satisfied: scipy $\geq$ 0.14 in ./anaconda3/lib/python3.8/site-packages (from Theano) (1.6.2)

Requirement already satisfied: six $\geq$ 1.9.0 in ./anaconda3/lib/python3.8/site-packages (from Theano) (1.15.0)

```
In [ ]: import theano.tensor as T
        from theano import function
```

```
In [ ]: # Declaring 2 variables
        x = T.dscalar('x')
        y = T.dscalar('y')
```

```
In [ ]: # Summing up the 2 numbers  
z = x + y
```

```
In [ ]: # Converting it to a callable object so that it takes matrix as parameters  
f = function([x, y], z)
```

```
In [ ]: f(5, 7)
```

```
Out[33]: array(12.)
```

4. PyTorch

```
In [ ]: !pip3 install torch torchvision torchaudio --extra-index-url  
https://download.pytorch.org/whl/cu115
```

```
Looking in indexes: https://pypi.org/simple,  
https://download.pytorch.org/whl/cu115  
Requirement already satisfied: torch in ./anaconda3/lib/python3.8/site-  
packages (1.12.1)  
Requirement already satisfied: torchvision in  
./anaconda3/lib/python3.8/site-packages (0.13.1)  
Requirement already satisfied: torchaudio in ./anaconda3/lib/python3.8/site-  
packages (0.12.1)  
Requirement already satisfied: typing-extensions in  
./anaconda3/lib/python3.8/site-packages (from torch) (3.7.4.3)  
Requirement already satisfied: requests in ./anaconda3/lib/python3.8/site-  
packages (from torchvision) (2.25.1)  
Requirement already satisfied: pillow $\neq$ 8.3.*,  $\geq$ 5.3.0 in  
./anaconda3/lib/python3.8/site-packages (from torchvision) (8.2.0)  
Requirement already satisfied: numpy in ./anaconda3/lib/python3.8/site-  
packages (from torchvision) (1.20.1)  
Requirement already satisfied: chardet $<$ 5,  $\geq$ 3.0.2 in  
./anaconda3/lib/python3.8/site-packages (from requests $\rightarrow$ torchvision) (4.0.0)  
Requirement already satisfied: idna $<$ 3,  $\geq$ 2.5 in  
./anaconda3/lib/python3.8/site-packages (from requests $\rightarrow$ torchvision) (2.10)  
Requirement already satisfied: urllib3 $<$ 1.27,  $\geq$ 1.21.1 in  
./anaconda3/lib/python3.8/site-packages (from requests $\rightarrow$ torchvision)  
(1.26.4)  
Requirement already satisfied: certifi $\geq$ 2017.4.17 in  
./anaconda3/lib/python3.8/site-packages (from requests $\rightarrow$ torchvision)  
(2020.12.5)
```

```
In [ ]: import torch  
import torch.nn as nn
```

```
In [ ]: print(torch.__version__)
```

```
1.12.1+cu102
```

```
In [ ]: torch.cuda.is_available()
```

```
Out[37]: False
```